

ADD R (Hex)	ADD R (Dec)	REGISTER NAME	FUNCTION	SERI AL I/F	Bit1 5	Bit1 4	Bit1 3	Bit1 2	Bit1 1	Bit1 0	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
00	00	SAVE	save/reboot/r eset	R/W	SAVE[15:0]															
01	01	CALSW	calibration mode	R/W																CALSW[3:0]
04	04	BAUD	Serial port baud rate	R/W																BAUD[3:0]
05	05	AXOFFSET	Acceleration X offset	R/W	AXOFFSET[15:0]															
06	06	AYOFFSET	Acceleration Y offset	R/W	AYOFFSET[15:0]															
07	07	AZOFFSET	Acceleration Z offset	R/W	AZOFFSET[15:0]															

08	08	GXOFFSET	Angular velocity X zero bias	R/W	GXOFFSET[15:0]
09	09	GYOFFSET	Angular velocity Y zero bias	R/W	GYOFFSET[15:0]
0A	10	GZOFFSET	Angular velocity Z zero bias	R/W	GZOFFSET[15:0]
0B	11	HXOFFSET	Magnetic Field X Zero Bias	R/W	HXOFFSET[15:0]
0C	12	HYOFFSET	Magnetic field Y zero bias	R/W	HYOFFSET[15:0]
0D	13	HZOFFSET	Magnetic field Z zero bias	R/W	HZOFFSET[15:0]
0E	14	WORKMODE E【1】	Z-axis operation mode	R/W	WORKMODE[15:0]
10	16	GYROPTP 【1】	Z-axis static peak-to-peak	R/W	GYROPTP[15:0]
11	17	GPTPTIME 【1】	Z-axis peak- to-peak acquisition time	R/W	GPTPTIME[15:0]
12	18	GYROBAIS 【1】	Z axis zero bias value	R/W	GYROBAIS[15:0]
13	19	GBAISTIME 【1】	Z axis zero bias acquisition time	R/W	GBAISTIME[15:0]

21	33	ACCRANGE	Acceleration range	RW															ACCRANGE[3:0]
22	34	SLEEP	sleep	RW															SLEEP
23	35	ORIENT	Installation direction	RW															ORIENT
24	36	AXIS6	algorithm	RW															AXIS6
25	37	FILTK	Dynamic filtering	RW	FILTK[15:0]														
26	38	GPSBAUD	GPS baud rate	RW															GPSBAUD[3:0]
27	39	READADDR	read register	RW										READADDR[7:0]					
2A	42	ACCFILT	acceleration filter	RW	ACCFILT[15:0]														
2E	46	VERSION	version number	R	VERSION[15:0]														
30	48	YYMM	year/month	RW	MOUTH[15:8]								YEAR[7:0]						
31	49	DDHH	day/hour	RW	HOUR[15:8]								DAY[7:0]						
32	50	MMSS	minutes/seconds	RW	SECONDS[15:8]								MINUTE[7:0]						
33	51	MS	millisecond	RW	MS[15:0]														
34	52	AX	acceleration X	R	AX[15:0]														
35	53	AY	acceleration Y	R	AY[15:0]														
36	54	AZ	acceleration	R	AZ[15:0]														

			Z		
37	55	GX	Angular velocityX	R	GX[15:0]
38	56	GY	Angular velocityY	R	GY[15:0]
39	57	GZ	Angular velocity Z	R	GZ[15:0]
3A	58	HX	Magnetic FieldX	R	HX[15:0]
3B	59	HY	Magnetic field Y	R	HY[15:0]
3C	60	HZ	Magnetic field Z	R	HZ[15:0]
3D	61	LRoll	X-axis angle low	R	LRoll[15:0]
3E	62	HRoll	X-axis angle high	R	HRoll[15:0]
3F	63	LPitch	Y-axis angle low	R	LPitch[15:0]
40	64	HPitch	Y-axis angle high	R	HPitch[15:0]
41	65	LYaw	Z-axis angle low	R	LYaw[15:0]
42	66	HYaw	Z-axis angle high	R	HYaw[15:0]
43	67	TEMP	Module temperature	R	TEMP[15:0]
61	97	GYROCALIT	Gyro	R/W	GYROCALITHR[15:0]

		HR	Calibration Threshold			
63	99	GYROCALTIME	Gyro calibration time	R/W	GYROCALTIME[15:0]	
69	105	KEY	unlock	R/W	KEY[15:0]	
6A	105	WERROR	Gyroscope change value	R/W	WERROR[15:0]	
6E	110	WZTIME	Z-axis still time threshold	R/W	WZTIME[15:0]	
6F	111	WZSTATIC	Z-axis static threshold	R/W	WZSTATIC[15:0]	
74	116	MODDELAY	485 data response delay	R/W	MODDELAY[15:0]	
7F	127	NUMBERID1	Device No.1- 2	R	ID2[15:8]	ID1[7:0]
80	128	NUMBERID2	Device No3-4	R	ID4[15:8]	ID3[7:0]
81	129	NUMBERID3	Device No5-6	R	ID6[15:8]	ID5[7:0]
82	130	NUMBERID4	Device No7-8	R	ID8[15:8]	ID7[7:0]
83	131	NUMBERID5	Device No9- 10	R	ID10[15:8]	ID9[7:0]
84	132	NUMBERID6	Device No11- 12	R	ID12[15:8]	ID11[7:0]
95	149	LREFROLL	ROLL angle zero is low	R/W	LREFROLL[15:0]	
96	150	HREFROLL	ROLL angle zero bias	R/W	HREFROLL[15:0]	

			high		
97	151	LREFPITCH	PITCH corner zero bias low	R/W	LREFPITCH[15:0]
98	152	HREFPITCH	PITCH corner zero bias high	R/W	HREFPITCH[15:0]

explanation:

1. The callout register is only applicable to HWTX073

Protocol format

Read register format

- Data is sent in hexadecimal, not ASCII.
- Each register address, the number of read registers, and the read data are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, the high and low bits of the number of registers to be read are represented by LEN_H and LEN_L, and the high and low bits of the read data are represented by DATA_{1H} and DATA_{1L}.
- The last two bits of the read command are standard CRC check bits. It can be calculated using the CRC check digit calculation tool.

command send

Modbus address	function code	registers high 8 bits	registers lower 8 bits	Read length high 8 bits	Read length lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (read)	ADDR _H [15:8]	ADDR _L [7:0]	LEN _H [15:8]	LEN _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus address	function code	read length	Data high 8 bits	Data lower 8 bits	Data high 8 bits	Data lower 8 bits	Data high 8 bits	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (read)	LENL[7:0]	DATA1H[15:8]	DATA1L[7:0]			DATANH	DATAL	CRCH[15:8]	CRCL[7:0]

Example: Read acceleration, angular velocity, magnetic field

Read register format

- Data is sent in hexadecimal, not ASCII.
- For each register address, the write data is represented by two bytes. The high and low bits of the register address are represented by ADDRH and ADDRL, and the high and low bits of the written data are represented by DATAH and DATAL.

command send

Modbus address	function code	register high 8 bits	register lower 8 bits	Data high 8 bits	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus address	function code	register high 8 bits	register lower 8 bits	Data high 8 bits	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Register Description

All the following examples are commands when the Modbus address is 0x50 (default). If you change the Modbus address, you need to change the address and CRC check bit in the command accordingly.

SAVE (save/reboot/reset)

Register Name: SAVE

Register Address: 0 (0x00)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Reboot: 0x00FF Factory reset: 0x0001

example:

send: 50 06 00 00 00 FF C4 0B (reboot)

return: 50 06 00 00 00 FF C4 0B

CALSW

Register Name: CALSW

Register Address: 1 (0x01)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	CAL[3:0]	To set the calibration mode: 0000(0x00): normal working mode 0001(0x01): Auto adder calibration 0011(0x03): height reset 0100(0x04): Set the heading angle to zero 0111(0x07): Magnetic Field Calibration (Spherical Fitting) 1000(0x08): set angle reference 1001(0x09): Magnetic Field Calibration (Dual Plane Mode)

example:

send: 50 06 00 01 00 04 D4 48 （Heading angle set to zero）

return: 50 06 00 01 00 04 D4 48

BAUD RATE

Register Name: BAUD RATE

Register Address: 4 (0x04)

Read and write direction: R/W

Default: 0x0002

Bit	NAME	FUNCTION
15:4		
3:0	BAUD[3:0]	Set serial port baud rate: 0001(0x01): 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps 0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps 1000(0x08): 460800bps (Only support WT931/JY931/HWT606/HWT906) 1001(0x09): 921600bps (Only support WT931/JY931/HWT606/HWT906)

example:

send: 50 06 00 04 00 06 45 88 (Set serial port baud rate 115200)

return: 50 06 00 04 00 06 45 88

AXOFFSET~HZOFFSET (Zero bias setting)

Register Name: AXOFFSET~HZOFFSET

Register Address: 5~13 (0x05~0x0D)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	AXOFFSET[15:0]	Acceleration X-axis offset, actual acceleration offset=AXOFFSET[15:0]/10000(g)
15:0	AYOFFSET[15:0]	Acceleration Y-axis offset, actual acceleration offset=AYOFFSET[15:0]/10000(g)
15:0	AZOFFSET[15:0]	Acceleration Z-axis offset, actual acceleration offset=AZOFFSET[15:0]/10000(g)
15:0	GXOFFSET[15:0]	Angular velocity X-axis offset, actual angular velocity offset=GXOFFSET[15:0]/10000(°/s)
15:0	GYOFFSET[15:0]	Angular velocity Y-axis offset, actual angular velocity offset=GYOFFSET[15:0]/10000(°/s)
15:0	GZOFFSET[15:0]	Angular velocity Z-axis offset, actual angular velocity offset=GZOFFSET[15:0]/10000(°/s)
15:0	HXOFFSET[15:0]	Magnetic field X-axis zero bias
15:0	HYOFFSET[15:0]	Magnetic field Y-axis zero bias
15:0	HZOFFSET[15:0]	Magnetic field Z-axis zero bias

example:

send: 50 06 00 05 03 E8 94 F4 (Set acceleration X-axis zero
bias 0.1g) , $0x03E8=1000$, $1000/10000=0.1(g)$

return: 50 06 00 05 03 E8 94 F4

WORKMODE (Z-axis operation mode)

Register Name: WORKMODE

Register address: 14(0x0E)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	WORKMODE[3:0]	Set the Z-axis operation mode: 0000(0x00): normal data mode 0001(0x01): peak-to-peak mode 0010(0x02): seek zero bias mode 0011 (0x03): find scale factor mode

example:

send: 50 06 00 0E 00 01 24 48 (Automatically obtain zero bias)

return: 50 06 00 0E 00 01 24 48

GYROPTP (Z-axis static peak-to-peak)

Register Name: GYROPTP

Register address: 16(0x10)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GYROPTP[15:0]	Parameters used in calculating zero offset, no need to set for sensor automatic acquisition

Write 0x01 to the "WORKMODE" register to enter the "peak-to-peak mode", in this mode the sensor automatically calculates the maximum and minimum values of the Z-axis angular velocity within the time set by "GPTPTIME" and records them in "GYROPTP". In "Zero Offset Mode", this data will be used to calculate and filter the zero offset.

Z-axis static peak-to-peak value = $\text{GYROPTP}/1000$ (°/s)

example:

send: 50 03 00 10 00 01 88 4E (read Z-axis static peak-to-peak)

return: 50 03 02 00 64 44 63

0x0064=100, Z-axis static peak-to-peak = $100/1000=0.1$ (°/s)

GPTPTIME （Z-axis peak-to-peak acquisition time）

Register name: GPTPTIME

Register address: 17(0x11)

read and write direction: R/W

Defaults: 0x000A

Bit	NAME	FUNCTION
15:0	GPTPTIME[15:0]	Calculate the peak-to-peak time, the default is 10S

Example:

Send: 50 06 00 11 00 0A 54 49 (set the Z-axis peak-to-peak acquisition time to 10S)

Return: 50 06 00 11 00 0A 54 49

After entering the "Peak-to-Peak Mode", obtain the difference between the maximum value and the minimum value of the Z-axis angular velocity within the "GPTPTIME" time, store it in "GYROPTP", and use this data for zero bias in the "Zero Bias Mode". calculation filter.

GYROBAIS (Z-axis zero bias value)

Register Name: GYROBAIS

Register address: 18(0x12)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GYROBAIS[15:0]	The zero bias value of the horizontal gyroscope, which can be obtained through the "Zero Bias Mode"

The Z-axis horizontal gyroscope has a certain zero bias value when placed at rest, and the angular velocity at rest can be eliminated by this zero bias value. The zero offset value can be automatically calculated by automatically obtaining the zero bias. Enter the "Zero bias Mode" to automatically calculate the zero bias value according to the peak-to-peak value of "GYROPTP" and the "GBAISTIME" bias acquisition time.

Z axis zero bias value = GYROBAIS/1000 (°/s)

example:

send: 50 03 00 12 00 01 88 4E (Read Z-axis static peak-to-peak value)

return: 50 03 02 00 64 B1 0F

0x0064=100, Z-axis static peak-to-peak value = 100/1000=0.1 (°/s)

GBAISTIME (Z-axis zero bias acquisition time)

Register Name: GBAISTIME

Register Address: 19(0x13)

Read and write direction: R/W

Default: 0x000A

Bit	NAME	FUNCTION
15:0	GBAISTIME[15:0]	Time required to calculate the zero offset value of the horizontal gyroscope

Example:

Send: 50 06 00 13 00 0A 2F5 89 (set the Z-axis zero bias acquisition time to 10S)

Return: 50 06 00 13 00 0A 2F5 89

The time required to obtain the Z-axis zero biast, and the zero bias value is obtained according to this time when the zero bias is obtained.

GSTATICTHRE (Z-axis static threshold)

Register Name: GSTATICTHRE

Register address: 20(0x14)

Read and write direction: R/W

Default: 0x0032

Bit	NAME	FUNCTION
15:0	GSTATICTHRE[15:0]	The larger the GSTATICTHRE[15:0], the weaker the seismic performance and the smaller the error The smaller the GSTATICTHRE[15:0], the better the seismic performance and the larger the error. Default value 50

The Z-axis horizontal gyroscope has slight data jitter when it is placed at rest. This parameter can filter out these slight jitters. When the angular velocity is less than the "GSTATICTHRE" setting value and continues for the time set by "GSTATICTIME", it is regarded as stationary, and the Z-axis angular velocity is zero. This parameter can be appropriately increased when it is used in scenes with jitter and the Z-axis is accumulated due to jitter, and this parameter can be appropriately decreased in scenes with very slow and uniform rotation.

Z-axis zero bias value = GSTATICTHRE/1000 (°/s)

example:

send: 50 06 00 14 00 64 C5 A4 ((set Z-axis static threshold0.1g) ,0x0064=100, 100/10000=0.1(°)

return: 50 06 00 14 00 64 C5 A4

GSTATICTIME (Z-axis stabilization time)

Register Name: GSTATICTIME

Register address: 21(0x15)

Read and write direction: R/W

Default: 0x0064

Bit	NAME	FUNCTION
15:0	GSTATICTIME[15:0]	Z-axis judges the still time

Z-axis static judgment time threshold. It is considered stationary when the angular velocity is less than the value set by "GSTATICTHRE" for the time set by "GSTATICTIME". If the stabilization time is required to be higher, this parameter can be adjusted appropriately. Lowering this parameter can speed up settling time but may also increase error.

Z-axis stabilization time = GSTATICTIME/1000 (s)

example:

send: 50 06 00 15 00 32 14 5A ((Set Z-axis stabilization time 0.05s) ,0x0032=50, 50/10000=0.05(°))

return: 50 06 00 15 00 32 14 5A

PGSCALE (Z-axis calibration factor P)

Register Name: PGSCALE		
Register address: 22(0x16)		
Read and write direction: R/W		
Default: 0x2710		
Bit	NAME	FUNCTION
15:0	PGSCALE[15:0]	range: 0~20000 This parameter is written in the factory using a high-precision turntable, please do not modify it
There is an error in the measurement of the Z-axis gyroscope, and a high-precision turntable will be used to measure this error when leaving the factory, and this parameter will be written into the sensor. Do not modify this parameter unless necessary. This parameter can be automatically calculated in the "scale factor mode". After entering the "scale factor mode", rotate the sensor by the angle set by "GSCALERANGE" to calculate the calibration factor. Z-axis calibration factor $P=PGSCALE/10000.0$ example: send: 50 03 00 16 00 01 68 4F (Read Z-axis calibration factor P) return: 50 03 02 27 74 24 48 0x2774 = 10100, Read Z-axis calibration factor $P=10100/10000=1.01$		

GSCALERANGE (Z-axis calibration angle)

Register Name: GSCALERANGE		
Register address: 24(0x18)		
Read and write direction: R/W		
Default: 0x02D0		
Bit	NAME	FUNCTION
15:0	GSCALERANGE[15:0]	When calibrating factors, it is necessary to calibrate according to this parameter
example:		
send: 50 06 00 18 01 68 04 32 ((Set the Z-axis calibration angle to 360) , 0x0168=360)		
return: 50 06 00 18 01 68 04 32		
The angle of rotation required in the Z-axis "scale factor mode", generally set to an integer multiple of 360°.		

IICADDR （Device address）

Register Name: IICADDR

Register Address: 26 (0x1A)

Read and write direction: R/W

Default: 0x0050

Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address for I2C and Modbus communication 0x01~0x7F

example:

send: 50 06 00 1A 00 02 24 4D （Set the device address to0x02）

return: 50 06 00 1A 00 02 24 4D

LEDOFF (turn off LED light)

Register Name: LEDOFF

Register Address: 27 (0x1B)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:1		
0	LEDOFF	1: turn off LED light 0: turn on LED light

example:

send: 50 06 00 1B 00 01 35 8C (turn off LED light)

return : 50 06 00 1B 00 01 35 8C

MAGRANGX~MAGRANGZ (Magnetic Field Calibration Range)

Register Name: MAGRANGX~MAGRANGZ

Register Address: 28~30 (0x1C~0x1E)

Read and write direction: R/W

Default: 0x01F4

Bit	NAME	FUNCTION
15:0	MAGRANGX[15:0]	Magnetic field calibration X-axis range
15:0	MAGRANGY[15:0]	Magnetic Field Calibration Y-axis Range
15:0	MAGRANGZ[15:0]	Magnetic field calibration Z-axis range

example:

send: 50 06 00 1C 01 F4 45 9A (Set the magnetic field calibration X-axis range to 500)

return: 50 06 00 1C 01 F4 45 9A

BANDWIDTH (bandwidth)

Register Name: BANDWIDTH		
Register Address: 31 (0x1F)		
Read and write direction: R/W		
Default: 0x0004		
Bit	NAME	FUNCTION
15:4		
3:0	BANDWIDTH[3:0]	set bandwidth 0000(0x00): 256Hz 0001(0x01): 188Hz 0010(0x02): 98Hz 0011(0x03): 42Hz 0100(0x04): 20Hz 0101(0x05): 10Hz 0110(0x06): 5Hz
Example: Send: 50 06 00 1F 00 01 74 4D (set the bandwidth to 188Hz) Return: 50 06 00 1F 00 01 74 4D		

GYRORANGE (Gyroscope range)

Register Name: GYRORANGE

Register Address: 32 (0x20)

Read and write direction: R/W

Default: 0x0003

Bit	NAME	FUNCTION
15:4		
3:0	GYRORANGE[3:0]	Set the gyro range 0011(0x03): 2000°/s The default is 2000°/s, fixed and cannot be set

Example:

Send: 50 06 00 20 00 03 C5 80 (set the gyro range to 2000°/s)

Return: 50 06 00 20 00 03 C5 80

ACCRANGE (Accelerometer range)

Register Name: ACCRANGE

Register Address: 33 (0x21)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	ACCRANGE[3:0]	Set the accelerometer range 0000(0x00): ±2g 0011(0x03): ±16g This parameter cannot be set. The internal adaptive acceleration range of the product will automatically switch to 16g when the acceleration exceeds 2g.

example:

send: 50 06 00 21 00 03 94 40 (Set the accelerometer range to 16g)

return: 50 06 00 21 00 03 94 40

SLEEP (sleep)

Register Name: SLEEP

Register Address: 34 (0x22)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:1		
0	SLEEP	set sleep 1(0x01): sleep Any serial port data, can wake up

example:

send: 50 06 00 22 00 01 E5 81 (enter sleep)

return: 50 06 00 22 00 01 E5 81

ORIENT （Installation direction）

Register Name: ORIENT

Register Address: 35 (0x23)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:1		
0	ORIENT	Set the installation direction 0 (0x00): horizontal installation 1(0x01): vertical installation (the Y-axis arrow of the coordinate axis must be upward)

example:

send: 50 06 00 23 00 01 B4 41 （Set up vertical installation）

return: 50 06 00 23 00 01 B4 41

AXIS6 （algorithm）

Register Name: AXIS6

Register Address: 36 (0x24)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:1		
0	AXIS6	set algorithm 0(0x00): 9-axis algorithm (magnetic field solution navigation angle, absolute heading angle) 1(0x01): 6-axis algorithm (integral solution navigation angle, relative heading angle)

Example:

Send: 50 06 00 24 00 01 05 80 (set the 6-axis algorithm mode)

Return: 50 06 00 24 00 01 05 80

FILTK (K value filter)

Register Name: FILTK		
Register Address: 37 (0x25)		
Read and write direction: R/W		
Default: 0x001E		
Bit	NAME	FUNCTION
15:0	FILTK[15:0]	Range: 1~10000, the default is 30 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 30) The smaller the FILTK[15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the FILTK[15:0], the weaker the seismic performance and the stronger the real-time performance.
example:		
send: 50 06 00 25 00 1E 15 88 (Set the K value filter to 30)		
return: 50 06 00 25 00 1E 15 88		

ACCFILT （acceleration filter）

Register Name: ACCFILT

Register Address: 42 (0x2A)

Read and write direction: R/W

Default: 0x01F4

Bit	NAME	FUNCTION
15:0	ACCFILT[15:0]	Range: 1~10000, the default is 500 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 500) The smaller the ACCFILT[15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the ACCFILT[15:0], the weaker the seismic performance and the stronger the real-time performance. This parameter is an empirical value, which needs to be debugged according to different environments. In the tractor environment, ACCFILT[15:0] can be adjusted to 100, because the vibration of the tractor is serious and the anti-vibration performance needs to be improved

Example:

Send: 50 06 00 2A 01 F4 A5 94 (set acceleration filter 500)

Return: 50 06 00 2A 01 F4 A5 94

VERSION (version number)

Register Name: VERSION

Register Address: 46 (0x2E)

Read and write direction: R

Default: none

Bit	NAME	FUNCTION
15:0	VERSION[15:0]	Different products, different version numbers

Example:

Send: 50 03 00 2E 00 01 E9 82 (read version number)

Return: 50 03 02 VH VL CRCH CRCL

VERSION[15:0]=(short)((((short)VH<<8)|VL)

YYMM~MS (on-chip time)

Register Name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Read and write direction: RW

Default: 0x0000

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	mouth
7:0	YYMM[7:0]	year
15:8	DDHH[15:8]	hour
7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	second
7:0	MMSS[7:0]	munite
15:0	MS[15:0]	millisecond

Example:

Send: 50 06 00 30 03 16 05 7A (set year 22-03)

Return: 50 06 00 30 03 16 05 7A

Send: 50 06 00 31 09 0C D3 D1 (set date 12-09)

Return: 50 06 00 31 09 0C D3 D1

Send: 50 06 00 32 3A 1E B7 2C (set minutes and seconds 30:58)

Return: 50 06 00 32 3A 1E B7 2C

Send: 50 06 00 33 01 F4 74 53 (set milliseconds 500)

Return: 50 06 00 33 01 F4 74 53

AX~AZ (acceleration)

Register Name: AX~AZ

Register address: 52~54 (0x34~0x36)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	acceleration $X=AX[15:0]/32768*16g$ (g is the gravitational acceleration, preferably 9.8m/s ²)
15:0	AY[15:0]	acceleration $Y=AY[15:0]/32768*16g$ (g is the gravitational acceleration, preferably 9.8m/s ²)
15:0	AZ[15:0]	acceleration $Z=AZ[15:0]/32768*16g$ (g is the gravitational acceleration, preferably 9.8m/s ²)

Example:

Send: 50 03 00 34 00 03 49 84 (read three-axis acceleration)

Return: 50 03 06 AXH AXL AYH AYL AZH AZL CRCH CRCL

$AX[15:0]=((\text{short})AXH \ll 8)|AXL;$

$AY[15:0]=((\text{short})AYH \ll 8)|AYL;$

$AZ[15:0]=((\text{short})AZH \ll 8)|AZL;$

GX~GZ （ Angular velocity ）

Register Name: GX~GZ

Register address: 55~57 (0x37~0x39)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GX[15:0]	Angular velocityX=GX[15:0]/32768*2000°/s
15:0	GY[15:0]	Angular velocityY=GY[15:0]/32768*2000°/s
15:0	GZ[15:0]	Angular velocityZ=GZ[15:0]/32768*2000°/s

Example:

Send: 50 03 00 37 00 03 B9 84 (read triaxial angular velocity)

Return: 50 03 06 GXH GXL GYH GYL GZH GZL CRCH CRCL

GX[15:0]=((short)GXH <<8)|GXL;

GY[15:0]=((short)GYH <<8)|GYL;

GZ[15:0]=((short)GZH <<8)|GZL;

HX~HZ (magnetic field)

Register name: HX~HZ

Register Address: 58~60 (0x3A~0x3C)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	HX[15:0]	magnetic field X=HX[15:0] (unit: LSB)
15:0	HY[15:0]	magnetic field Y=HY[15:0] (unit: LSB)
15:0	HZ[15:0]	magnetic field Z=HZ[15:0] (unit : LSB)

example:

send: 50 03 00 3A 00 03 28 47 (Read the three-axis magnetic field)

return: 50 03 06 HXH HXL HYH HYL HZH HZL CRCH CRCL

$HX[15:0] = ((\text{short})HXH \ll 8) | HXL;$

$HY[15:0] = ((\text{short})HYH \ll 8) | HYL;$

$HZ[15:0] = ((\text{short})HZH \ll 8) | HZL;$

Roll~Yaw （angle）

Register Name: Roll~Yaw		
Register address: 61~66 (0x3D~0x42)		
Read and write direction: R		
Default: 0x0000		
Bit	NAME	FUNCTION
15:0	LRoll[15:0]	Roll Angle X Low
15:0	HRoll[15:0]	Roll angle X high position, roll angle $X = (((\text{int})\text{HRoll} \ll 16) \text{LRoll}) / 1000^\circ$
15:0	LPitch[15:0]	Pitch angle Y low
15:0	HPitch[15:0]	Pitch angle Y high position, pitch angle $X = (((\text{int})\text{HPitch} \ll 16) \text{LPitch}) / 1000^\circ$
15:0	LYaw[15:0]	Heading Angle Z Low
15:0	HYaw[15:0]	Heading angle Z high position, heading angle $Z = (((\text{int})\text{HYaw} \ll 16) \text{LYaw}) / 1000^\circ$
example:		
send: 50 03 00 3D 00 06 59 85 （Read the 3-axis angle）		
return: 50 03 0C LRollH LRollL HRollH HRollL LPitchH LPitchL HPitchH HPitchL LYawH LYawL HYawH HYawL CRCH CRCL		
roll angle （x-axis） $\text{Roll} = ((\text{HRollH} \ll 24) (\text{HRollL} \ll 16) (\text{LRollH} \ll 8) \text{LRollL}) / 1000^\circ$		
pitch angle （y-axis） $\text{Pitch} = ((\text{HPitchH} \ll 24) (\text{HPitchL} \ll 16) (\text{LPitchH} \ll 8) \text{LPitchL}) / 1000^\circ$		

yaw angle (z-axis)

$$\text{Yaw} = ((\text{HYawH} < 24) | (\text{HYawL} < 16) | (\text{LYawH} < 8) | \text{LYawL}) / 1000^\circ$$

TEMP (temperature)

Register Name: TEMP

Register Address: 67 (0x43)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	temperature=TEMP[15:0]/100°C

example:

send: 50 03 00 67 00 01 38 54 (Read Triaxial Magnetic Fields)

return: 50 03 02 TEMPH TEMPL CRCH CRCL

TEMP[15:0]=((short)TEMPH <<8)|TEMPL ;

q0~q3 (Quaternion)

Register Name: q0~q3

Register Address: 81~84 (0x51~0x54)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	q0[15:0]	Quaternion0=q0[15:0]/32768
15:0	q1[15:0]	Quaternion1=q1[15:0]/32768
15:0	q2[15:0]	Quaternion2=q2[15:0]/32768
15:0	q3[15:0]	Quaternion3=q3[15:0]/32768

example:

send: 50 03 00 51 00 04 18 59 (read quaternion)

return: 50 03 08 q0H q0L q1H q1L q2H q2L q3H q3L CRCH CRCL

q0[15:0]=((short)q0H <<8)|q0L ;

q1[15:0]=((short)q1H <<8)|q1L ;

q2[15:0]=((short)q2H <<8)|q2L ;

q3[15:0]=((short)q3H <<8)|q3L ;

GYROCALITHR (Gyro Still Threshold)

Register Name: GYROCALITHR

Register Address: 97 (0x61)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	GYROCALITHR[15:0]	set Gyro Still Threshold: Gyro Still Threshold value=GYROCALITHR[15:0]/1000(°/s)

example:

send: 50 06 00 61 00 32 54 40 (Set the gyro static threshold to 0.05°/s)

return: 50 06 00 61 00 32 54 40

When the angular velocity change is less than 0.05°/s and lasts for the time of "GYROCALTIME", the sensor recognizes it as stationary and automatically resets the angular velocity less than 0.05°/s to zero

The setting rule of the static threshold of the gyroscope can be determined by reading the value of the "WERROR" register. The general setting rule is:

$GYROCALITHR = WERROR * 1.2$, unit: °/s.

This register needs to be used in conjunction with the GYROCALTIME register

GYROCALTIME (Gyro auto calibration time)

Register Name: GYROCALTIME

Register Address: 99 (0x63)

Read and write direction: R/W

Default: 0x03E8

Bit	NAME	FUNCTION
15:0	GYROCALTIME[15:0]	set gyro auto calibration time

example:

send: 50 06 00 63 01 F4 74 42 (Set gyroscope auto-calibration time500ms,0x01F4=500(ms))

return: 50 06 00 63 01 F4 74 42

When the angular velocity change is less than "GYROCALITHR" and lasts for 500ms, the sensor recognizes that it is stationary and automatically resets the angular velocity less than 0.05°/s to zero

This register needs to be used in conjunction with the GYROCALITHR register

KEY (unlock)

Register Name: KEY

Register Address: 105 (0x69)

Read and write direction: R/W

Default: 0x0000

Bit	NAME	FUNCTION
15:0	KEY[15:0]	Unlock register: When performing a write operation, you need to set this register first

Example:

Send: 50 06 00 69 B5 88 22 A1 (unlocked)

Return: 50 06 00 69 B5 88 22 A1

Unlock, write 0xB588 to this register (other values are invalid)

WERROR (Gyroscope change value)

Register Name: WERROR

Register Address: 106 (0x6A)

Read and write direction: R

Default: 0x0000

Bit	NAME	FUNCTION
15:0	WERROR[15:0]	Gyroscope change value=WERROR[15:0]/1000*180/3.1415926(°/s) When the sensor is stationary, the "GYROCALITHR" register can be set by changing this register

WZTIME (Angular velocity continuous rest time)

Register Name: WZTIME

Register Address: 110 (0x6E)

Read and write direction: R/W

Default: 0x01F4

Bit	NAME	FUNCTION
15:0	WZTIME[15:0]	Angular velocity continuous rest time

Example:

Send: 50 06 00 6E 01 F4 E5 81 (set the angular velocity continuous static time 500ms, 0x01F4=500(ms))

Return: 50 06 00 6E 01 F4 E5 81

When the angular velocity is less than "WZSTATIC" and lasts for 500ms, the angular velocity output is 0, and the Z-axis heading angle is not integrated

This register needs to be used in conjunction with the "WZSTATIC" register

WZSTATIC (Angular velocity integral threshold)

Register Name: WZSTATIC

Register Address: 111 (0x6F)

Read and write direction: R/W

Default: 0x012C

Bit	NAME	FUNCTION
15:0	WZSTATIC[15:0]	Angular velocity integral threshold=WZSTATIC[15:0]/1000(°/s)

Example:

Send: 50 06 00 6F 01 F4 E5 81 (set the angular velocity integration threshold to 0.5°/s, 0x01F4=500, 500/1000=0.5(°/s))

Return: 50 06 00 6F 01 F4 E5 81

When the angular velocity is greater than 0.5°/s, the Z-axis heading angle starts to integrate the acceleration

When the angular velocity is less than 0.5°/s, and the duration set by the register "WZTIME", the angular velocity output is 0, and the Z-axis heading angle is not integrated

This register needs to be used in conjunction with the "WZTIME" register

MODDELAY (485 data response delay)

Register Name: MODDELAY

Register Address: 116 (0x74)

Read and write direction: R/W

Default: 0x0BB8

Bit	NAME	FUNCTION
15:0	MODDELAY[15:0]	Set 485 data response delay, default 3000, unit: us

Example:

Send: 50 06 00 74 03 E8 C4 EF (set 485 data response delay 1000us)

Return: 50 06 00 74 03 E8 C4 EF

When the sensor receives the Modbus read command, the sensor delays 1000us and returns data

This register only supports Modbus version of the sensor

NUMBERID1~NUMBERID6 (device number)

Register Name: NUMBERID1~NUMBERID6		
Register address: 127~132 (0x7F~0x84)		
Read and write direction: R		
Default: none		
Bit	NAME	FUNCTION
15:0	NUMBERID1[15:0]	
15:0	NUMBERID2[15:0]	
15:0	NUMBERID3[15:0]	
15:0	NUMBERID4[15:0]	
15:0	NUMBERID5[15:0]	
15:0	NUMBERID6[15:0]	
Device number: WT4200000001		

XREFROLL~YREFPITCH (Angle zero reference value)

Register Name: LREFROLL~HREFPITCH

Register address: 149~152 (0x95~0x98)

Read and write direction: R/W

Default: 0x00000

Bit	NAME	FUNCTION
15:0	LREFROLL[15:0]	Roll angle zero reference value low
15:0	HREFROLL[15:0]	Roll Angle Zero Reference Value High, Roll Angle Zero Reference =
15:0	LREFPITCH[15:0]	$((\text{int})\text{HREFROLL} \ll 16) \text{LREFROLL} / 1000(^{\circ})$
15:0	HREFPITCH[15:0]	Pitch angle zero reference value low

Example: The current roll angle is 2°, set the roll angle zero, subtract 2°, then
REFROLL[31:0]=2*1000=2000=0x07D0

LREFROLL[15:0]=0x07D0, HREFROLL[15:0]=0;

Send: 50 06 00 95 07 D0 97 CB (set "LREFROLL")

Return: 50 06 00 95 07 D0 97 CB

Send: 50 06 00 96 00 00 64 67 (set "HREFROLL")

Return: 50 06 00 96 00 00 64 67